

```
a=list(range(20))
%timeit [i**2 for i in a]
```

1.27 μ s $\pm$ 346 ns per loop (mean $\pm$ std. dev. of 7 runs, 1000000 loops each)

```
a=list(range(20000000))
%timeit [i**2 for i in a]
```

1.94 s $\pm$ 357 ms per loop (mean $\pm$ std. dev. of 7 runs, 1 loop each)

```
import numpy as np
b=np.array(range(20000000))
%timeit b**2
```

54.9 ms $\pm$ 569 μ s per loop (mean $\pm$ std. dev. of 7 runs, 10 loops each)

```
a=list(range(20))
type(a)
```

list

```
a=list(range(20000000))
type(b)
```

numpy.ndarray

```
b=np.array(range(20000000))
type(b)
b
```

array([0, 1, 2, ..., 19999997, 19999998, 19999999])

```
c1=np.arange(1,10)
print(c1)
```

[1 2 3 4 5 6 7 8 9]

```
c3=np.arange(1,10,0.5)
print(c3)
```

[1. 1.5 2. 2.5 3. 3.5 4. 4.5 5. 5.5 6. 6.5 7. 7.5 8. 8.5 9. 9.5]

```
b.ndim
```

1

```
len(b)
```

20000000

```
b.shape
```

(20000000,)

```
a=np.arange(1,10)
print(a)
print(a[-2])
```

[1 2 3 4 5 6 7 8 9]
8

```
a[2:5]
```

array([3, 4, 5])

```
a[3:]
```

array([4, 5, 6, 7, 8, 9])

```
#multi-indexing
a[[2,5]]
```

```
array([3, 6])
```

```
#multi-indexing Numpy  
a[[2,5,7,15]]
```

```
-----  
TypeError                                Traceback (most recent call last)  
/tmp/ipykernel_466/1996792270.py in <cell line: 0>()  
      1 #multi-indexing Numpy  
----> 2 a[[2,5,7,15]]  
  
TypeError: list indices must be integers or slices, not list
```

```
a=np.array([6,9,3,5,7,2,8,7])  
a<6
```

```
array([False, False,  True,  True, False,  True, False, False])
```

```
a[a<6]
```

```
array([3, 5, 2])
```

```
a[a%2==0]
```

```
array([6, 2, 8])
```

```
a=np.array([-6,9,-3,0,5,7,-2,8,1])  
print(a[a>0])
```

```
[9 5 7 8 1]
```

```
a=np.array(range(16))  
a
```

```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15])
```

```
a.shape
```

```
(16,)
```

```
b=a.reshape(4,4)  
print(b)
```

```
[[ 0  1  2  3]  
 [ 4  5  6  7]  
 [ 8  9 10 11]  
 [12 13 14 15]]
```

```
a=np.array(range(16))
```

```
b.reshape(4,-1)
```

```
array([[ 0,  1,  2,  3],  
       [ 4,  5,  6,  7],  
       [ 8,  9, 10, 11],  
       [12, 13, 14, 15]])
```

```
b.reshape(-1,4)
```

```
array([[ 0,  1,  2,  3],  
       [ 4,  5,  6,  7],  
       [ 8,  9, 10, 11],  
       [12, 13, 14, 15]])
```

```
a=np.arange(12).reshape(3,4)  
a
```

```
array([[ 0,  1,  2,  3],  
       [ 4,  5,  6,  7],  
       [ 8,  9, 10, 11]])
```

```
a.shape
```

```
(3, 4)
```

```
b=a.T
```

```
b
```

```
array([[ 0,  4,  8],  
       [ 1,  5,  9],  
       [ 2,  6, 10],  
       [ 3,  7, 11]])
```

```
b.ndim
```

```
2
```

```
b.shape
```

```
(4, 3)
```

```
a=np.arange(1,10).reshape(3,3)
```

```
a
```

```
array([[1, 2, 3],  
       [4, 5, 6],  
       [7, 8, 9]])
```

```
print(a)
```

```
print(a[1][2])
```

```
[[1 2 3]  
 [4 5 6]  
 [7 8 9]]
```

```
6
```

```
#2D slicing
```

```
a=np.arange(1,10)
```

```
print(a)
```

```
a[3:6]
```

```
[1 2 3 4 5 6 7 8 9]  
array([4, 5, 6])
```

```
#2D slicing
```

```
a[:2]
```

```
array([1, 2])
```

```
#2D slicing
```

```
a[:, :2]
```

```
a[:, 1:3]
```

```
-----  
NameError                                Traceback (most recent call last)  
/tmp/ipykernel_916/2672378787.py in <cell line: 0>()  
      1 #2D slicing  
----> 2 a[:, :2]  
      3 a[:, 1:3]  
  
NameError: name 'a' is not defined
```

Next steps: [Explain error](#)

```
a[1:,1:2]
```

```
-----  
TypeError                                Traceback (most recent call last)  
/tmp/ipykernel_466/2697409243.py in <cell line: 0>()  
----> 1 a[1:,1:2]  
  
TypeError: list indices must be integers or slices, not tuple
```

```
import numpy as np
a=np.array(range(25)).reshape(5,5)
print(a)
print("The element of that index",a[[0,2,4,3],[1,3,0,4]])
```

```
[[ 0  1  2  3  4]
 [ 5  6  7  8  9]
 [10 11 12 13 14]
 [15 16 17 18 19]
 [20 21 22 23 24]]
The element of that index [ 1 13 20 19]
```

```
mask=(a%2==0)
a[mask]=-1
print(a)
```

```
-----
NameError                                Traceback (most recent call last)
/tmp/ipykernel_466/1904918212.py in <cell line: 0>()
----> 1 mask=(a%2==0)
      2 a[mask]=-1
      3 print(a)

NameError: name 'a' is not defined
```

#Aggregate Function

```
a=np.array([10,20,35,40,55])
print(np.sum(a))
print(np.mean(a))
print(np.min(a))
print(np.max(a))
```

```
160
32.0
10
55
```

```
a=np.array([10,20,35,4,55])
np.all(a<50)
```

```
np.False_
```

```
np.any(a<30)
```

```
np.True_
```

```
a=np.array([10,20,35,40,55])
b=np.where(a%2!=0,-1,1)
```

```
b
```

```
array([ 1,  1, -1,  1, -1])
```

```
a=np.array([10,20,35,40,55])
b=np.where(a%2!=0,-10,10)
```

```
b
```

```
array([ 10,  10, -10,  10, -10])
```

```
#sorting
a=np.array([100,210,50,140,155])
b=np.sort(a)
b
```

```
array([ 50, 100, 140, 155, 210])
```

```
#sorting
a=np.array([100,210,50,140,155])
b=np.sort(a)
a
```

```
array([100, 210,  50, 140, 155])
```

```
a=np.array([100,210,50,140,155])
a.sort()
b=a.argsort()
```

```
a=np.arange(5)
a=np.ones(5)
a=np.zeros(5)
a
```

```
array([0., 0., 0., 0., 0.])
```

```
a=np.arange(1,13).reshape(3,4)
b=np.arange(2,14).reshape(3,4)
print(a,"\n",b)
```

```
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
[[ 2  3  4  5]
 [ 6  7  8  9]
 [10 11 12 13]]
```

```
a=np.arange(1,13).reshape(3,4)
b=np.arange(2,14).reshape(4,3)
print(a,"\n\n",b)
```

```
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
```

```
[[ 2  3  4]
 [ 5  6  7]
 [ 8  9 10]
 [11 12 13]]
```

```
#matrix multiplication
c=np.dot(a,b)
c
```

```
array([[ 80,  90, 100],
       [184, 210, 236],
       [288, 330, 372]])
```

```
#multiplication method
c=a@b
c
```

```
array([[ 80,  90, 100],
       [184, 210, 236],
       [288, 330, 372]])
```

```
c=np.matmul(a,b)
```

```
a=np.arange(5)
```

```
a
```

```
array([0, 1, 2, 3, 4])
```

```
a.ndim
```

```
1
```

```
a.shape
```

```
(5,)
```

```
b=np.arange(1,6)
b
```

```
array([1, 2, 3, 4, 5])
```

```
a=np.arange(5)
a
```

```
array([0, 1, 2, 3, 4])
```

```
c=np.dot(a,b)  
c
```

```
np.int64(40)
```